Problem Definition & Mathematical Model
Smart Ambulance Allocation System

1. Problem Definition

The problem is to allocate ambulances from a set of ambulance stations to a set of service areas while satisfying the available supply at every station and the required demand in every service area. The allocation should minimize the total estimated travel-time cost.

This is modeled as a transportation optimization problem. The input cost matrix gives the estimated travel time (in minutes) for assigning an ambulance from each station to each area. The model is a static planning model; it does not itself dispatch ambulances to live incidents or account dynamically for traffic and changing incident locations.

2. Sets and Indices

- i = 1, ..., 8: ambulance stations, denoted by S1–S8.
- j = 1, ..., 12: service areas, denoted by A1–A12.

3. Parameters

- s_i: available ambulance supply at station i.
- d_j: ambulance demand in service area j.
- c_ij: estimated travel-time cost in minutes from station i to area j.
- A closed road makes the corresponding assignment unavailable.
- A route capacity specifies the maximum number of ambulances that may be assigned on that station–area route.

4. Decision Variables

x_ij is the number of ambulances assigned from station i to service area j.

Because ambulances are countable, the decision variables must be nonnegative integers:

$x_{ij} \in \{0, 1, 2, ...\}$ for all i, j.

5. Mathematical Model

Objective function

Minimize the total estimated allocation cost:

$\text{Minimize } Z = \Sigma_{(i=1..8)} \Sigma_{(j=1..12)} c_{ij} x_{ij}$

Since c_ij is measured in minutes per assigned ambulance in this dataset, Z is a weighted allocation-cost total (ambulance-minutes), not the actual average response time of future incidents.

Supply constraints

Every station's available supply is allocated:

$\Sigma(j=1..12)\ x_{ij} = s_i$ for each station $i = 1, ..., 8$.

Demand constraints

Each service area's demand is met:

$\Sigma(i=1..8)\ x_{ij} = d_j$ for each area $j = 1, ..., 12$.

Closed-road constraint

The dataset marks the route from station S2 to area A7 as closed due to maintenance. Therefore:

$x_{(S2,A7)} = 0$.

Route-capacity constraint

The dataset limits the route from station S5 to area A9 to at most 5 ambulances:

$0 \le x_{(S5,A9)} \le 5$.

Integrality and nonnegativity

$x_{ij} \ge 0$ and integer for all i, j.

Supply-demand balance

For a balanced transportation model:

$\Sigma(i=1..8)\ s_i = \Sigma(j=1..12)\ d_j$.

For this dataset, both totals equal 78, so the equality supply and demand formulation is consistent.

6. Dataset

Supply by station

Station	Supply (ambulances)
S1	10
S2	10
S3	10
S4	10
S5	9
S6	10
S7	9
S8	10

Demand by service area

Measure	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
Demand	4	6	7	4	7	8	7	5	12	6	8	4

Estimated travel-time cost matrix (minutes)

Station	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
S1	17	24	31	15	47	43	32	14	11	42	33	31
S2	24	6	15	33	24	16	5	20	26	20	43	28
S3	31	34	40	8	56	53	26	25	17	47	28	31

S4	31	17	10	28	28	25	25	13	19	18	28	13
S5	37	25	9	33	23	26	30	24	1	13	29	14
S6	30	17	9	38	17	17	40	25	28	8	38	22
S7	33	22	8	33	20	20	30	24	23	12	29	14
S8	15	19	25	21	39	34	32	9	13	33	36	26

7. Solution Methodology

7.1 Vogel's Approximation Method (VAM)

VAM is commonly used to obtain an initial feasible solution for a transportation problem:

1. For every eligible row and column, calculate the penalty as the difference between its two smallest available costs.
2. Select the eligible row or column with the largest penalty.
3. In that row or column, allocate as much as possible to the eligible cell with the lowest cost, subject to remaining supply, remaining demand, and any route upper bound.
4. Exclude closed routes from consideration. Respect route capacities when making allocations.
5. Update supply and demand and repeat until all constraints are satisfied.

Important implementation note: Because this dataset includes both a prohibited route and a route upper bound, a standard textbook VAM implementation that only handles ordinary supply/demand constraints is not sufficient without modification. The implementation must exclude S2–A7 and enforce the S5–A9 upper bound.

7.2 MODI (Modified Distribution Method)

For a standard transportation basis, calculate row and column potentials u_i and v_j for basic cells using:

$u_i + v_j = c_{ij}$.

For an eligible non-basic cell, define the reduced cost as:

$\Delta_{ij} = c_{ij} - (u_i + v_j)$.

With this sign convention, for a minimization transportation problem, all eligible non-basic cells having $\Delta_{ij} \geq 0$ indicates optimality for the standard transportation model. A negative reduced cost indicates a possible improving entry; construct a valid closed loop, alternate plus/minus positions, and adjust by the smallest allocation at minus positions.

Important implementation note: The upper-bound constraint on S5–A9 makes this a bounded transportation problem. The MODI implementation must also account for a route at its upper bound (and must not treat a closed route as an ordinary eligible cell). For a rigorous constrained optimum, use a bounded transportation algorithm or a linear/integer optimization solver and verify all constraints. Do not apply the basic MODI stopping rule while ignoring the route bound.

8. Reference Solution for Validation

The file provides a reference optimal objective value of 1111 with constraints and 1016 without constraints. Solving the supplied cost-minimization model with the closed-route and route-capacity constraints gives objective value 1111, matching the constrained reference value.

The supplied current allocation has objective value 2097. It satisfies the supply and demand totals, but its objective value is higher than the constrained optimum.

Optimal allocation matrix (reference check)

Station	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
S1	0	0	0	0	0	0	0	0	7	0	3	0
S2	0	6	0	0	0	4	0	0	0	0	0	0
S3	0	0	0	4	0	0	1	0	0	0	5	0
S4	0	0	1	0	0	0	5	0	0	0	0	4
S5	0	0	4	0	0	0	0	0	5	0	0	0
S6	0	0	0	0	0	4	0	0	0	6	0	0
S7	0	0	2	0	7	0	0	0	0	0	0	0
S8	4	0	0	0	0	0	1	5	0	0	0	0

Objective value: Z = 1111.

This allocation satisfies all station supplies, all area demands, the closed-road restriction, and the S5–A9 route capacity. Use it as a validation target for the program, while making sure the chosen VAM/MODI implementation correctly handles the extra route constraints.

9. Assumptions and Limitations

6. The dataset values are synthetic as stated in the JSON file; they are not confirmed field measurements.
7. The cost matrix is treated as fixed during the planning period.
8. Supply and demand values are known in advance.
9. The model uses a static allocation and does not dynamically re-optimize after each live incident.
10. The objective is a sum of allocation costs, not a direct model of incident probabilities, clinical severity, or the average response time of actual calls.
11. Road closure and route capacity are explicitly represented, but other real-world restrictions are not included unless added as constraints.

10. Conclusion

The model minimizes total estimated allocation cost while satisfying station supply, service-area demand, route availability, route capacity, and integer allocation requirements. VAM can provide an initial solution if adapted to the route restrictions; MODI or an appropriate bounded optimization method can then be used to improve or verify the solution. The constrained reference objective is 1111 for the supplied dataset.